速通期中考

注意事項

- 考試時間：10/27 (18:10 ~ 20:10)
- 考試範圍：先前上課內容
- 規則：

    -- 繳交程式碼(.m檔)至 NCU ee-class

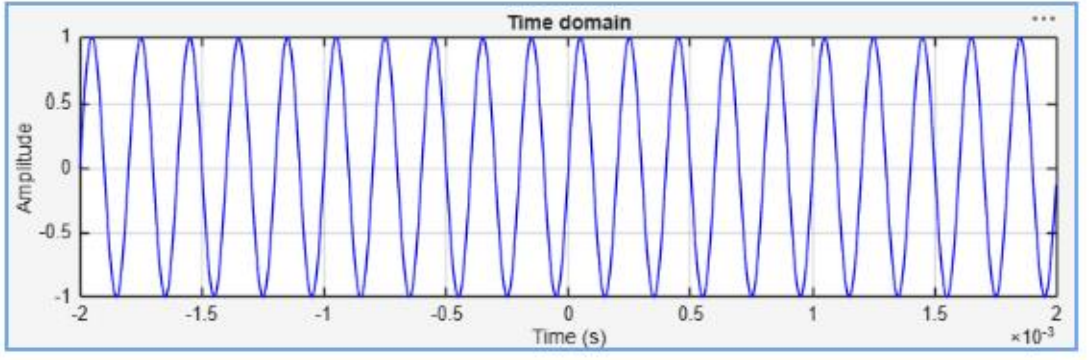
        檔名：你的學號_test1

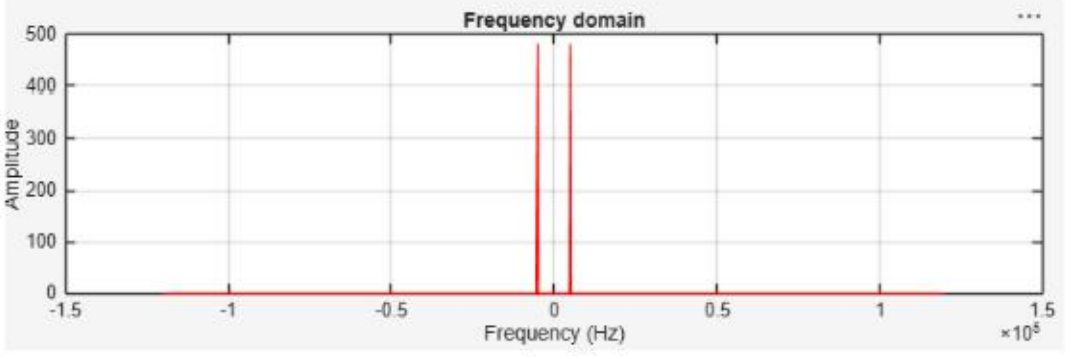
    -- 可使用作業和上課程式碼

觀念釐清 part 1

```
% 原始訊號
y = sin(2*pi*f0*t);
```

Fourier transform : t (s) ↔ f (Hz)






```
% 建立座標軸 (time domain)
fs = 240e3;
dt = 1/fs;
t = -0.002 : dt : 0.002-dt;
```

```
% 建立座標軸 (frequency domain)
N = length(y);
f = (-N/2:N/2-1)*fs/N;
```

```
% 建立座標軸 (time domain)
fs = 240e3;
dt = 1/fs;
t = -0.002 : dt : 0.002-dt;
```

Q：為什麼要 - dt？

A：讓等等的 N 比較好 / 2



-0.002



-0.002

不取終點

在資料點數量很多的情況下

對連續訊號沒有差別

```
% 建立座標軸 (frequency domain)
N = length(y);
f = (-N/2:N/2-1)*fs/N;
```

Workspace

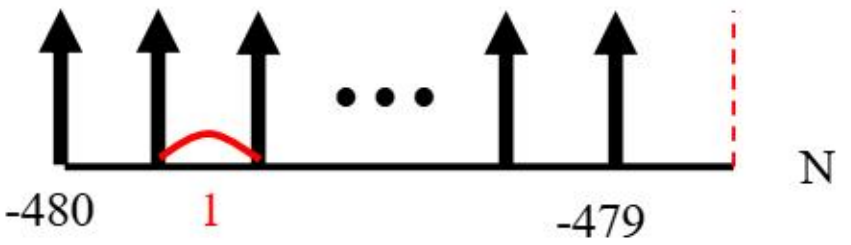
Name	Value	Size
f	1×960 double	1×960
f0	5000	1×1
fs	240000	1×1
N	960	1×1
t	1×960 double	1×960
y	1×960 double	1×960
Y	1×960 complex...	1×960

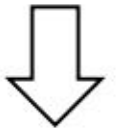
t → y → N

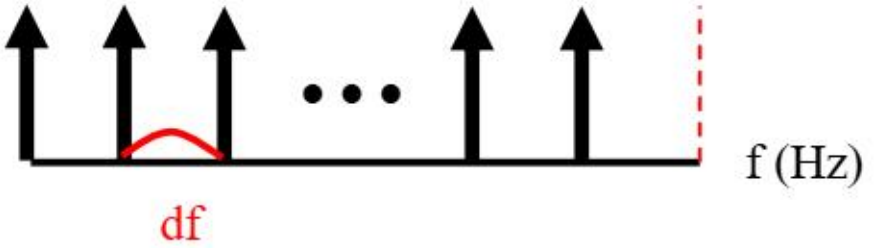
代表有 960 資料點數量

Q：為什麼要 / 2？

A：讓圖形對稱且 0 在正中間







df = fs / N

觀念釐清 part 2

```
% 原始訊號經過 LPF
y_lpf = zeros(length(fs_num)+1, length(t)); ---------- ①
y_lpf(1,:) = conv(y, h_lpf, 'same'); ---------- ②
```

①

建立都為零的矩陣

Row 1 > 0 0 0 0 0 0 ……共有 length(t) 個

Row 2 > 0 0 0 0 0 0 ……共有 length(t) 個

~

Row 6 > 0 0 0 0 0 0 ……共有 length(t) 個

②

回傳數值至矩陣的 Row1 中

Row 1 > 8 9 8 7 6 5 ……共有 length(t) 個

Row 2 > 0 0 0 0 0 0 ……共有 length(t) 個

~

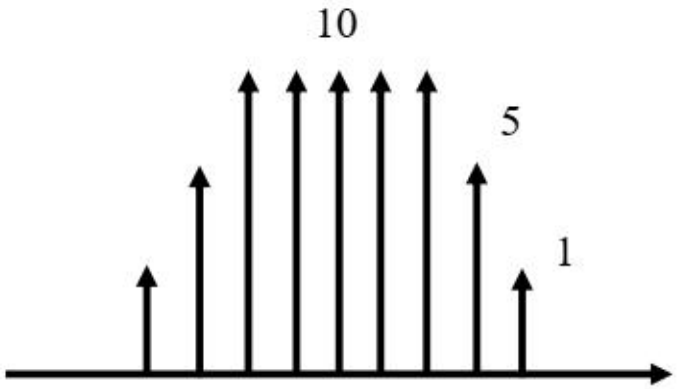
Row 6 > 0 0 0 0 0 0 ……共有 length(t) 個

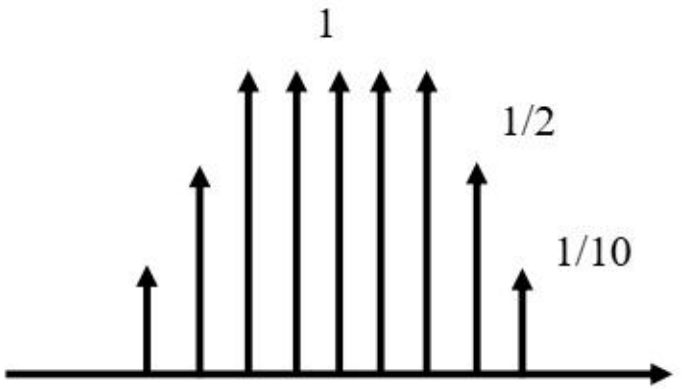
觀念釐清 part 3

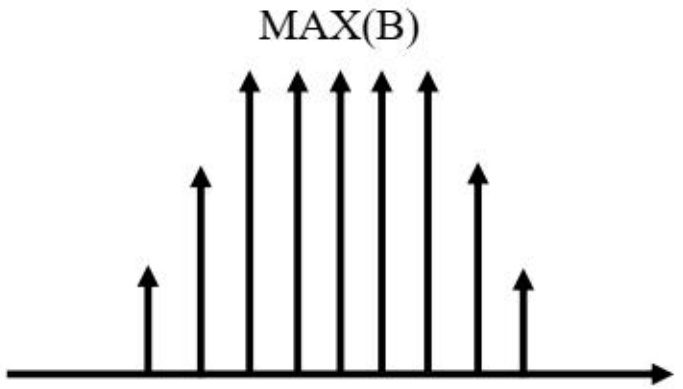
```
% 取樣後頻譜 + LPF
subplot(length(fs_num)+1,4,4*i+2)
plot(f, abs(fftshift(fft(y_sample(i,:)))), 'r', 'LineWidth', 1)
hold on
plot(f, abs(fftshift(fft(h_lpf)))./max(abs(fftshift(fft(h_lpf)))) .* max(abs(fftshift(fft(y_sample(i,:))))), 'k', 'LineWidth', 1)
grid on
xlim(6*[-f0 f0])
xlabel('f (Hz)')
ylabel('|X_s(f)| & |H_{LPF}(f)|')
title('Freq. Response of Sampled Signal and LPF')
hold off
```

A ./ MAX(A) .*MAX(B)

為了讓輸出圖形好看 (取完整訊號)







重點整理 ▸ Nyquist sampling

原始訊號

⬇

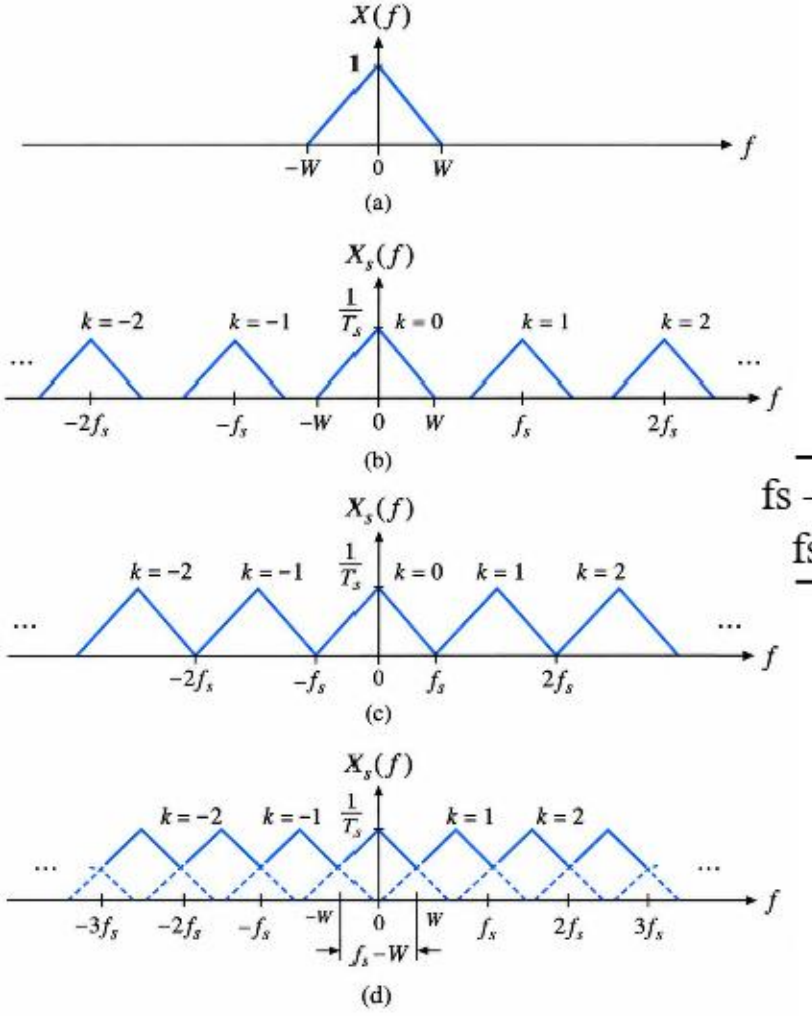
取樣後

⬇

fs 降低

⬇

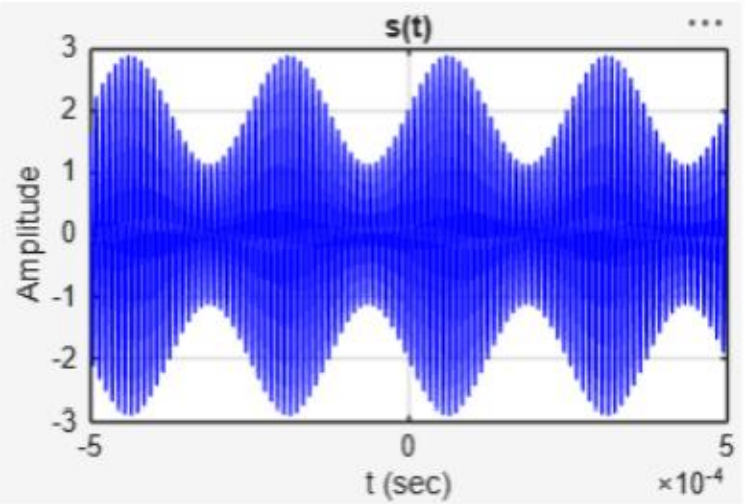
fs 再降低

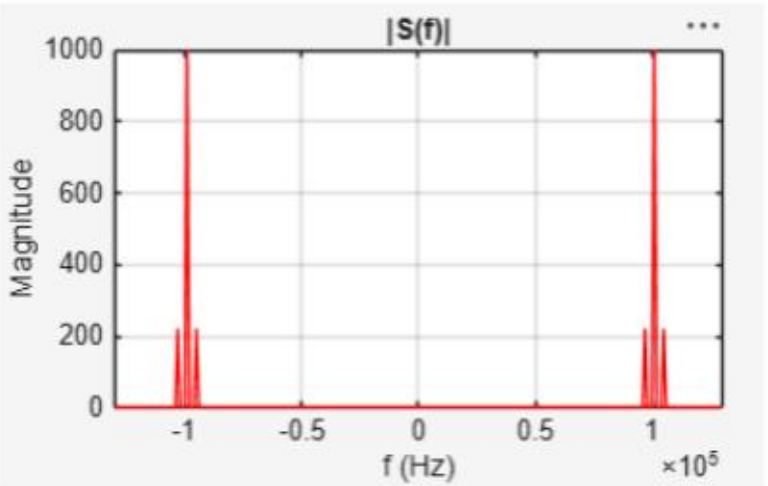


fs – W > W
fs > 2W

Aliasing !!!

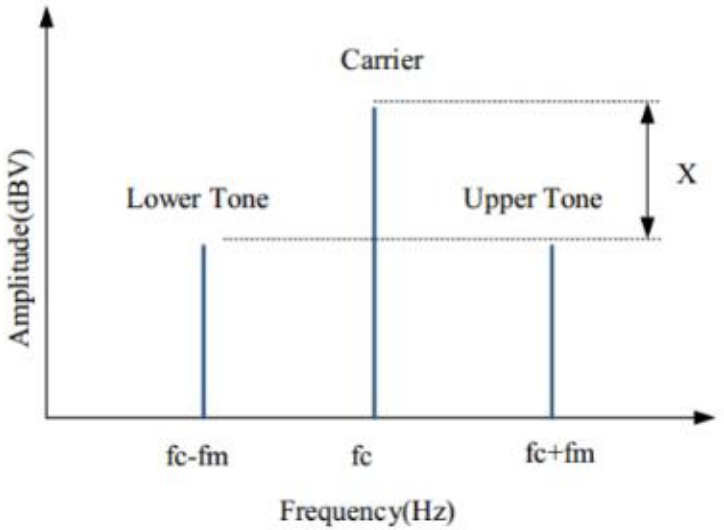
重點整理 ▸ AM modulation (Amplitude)



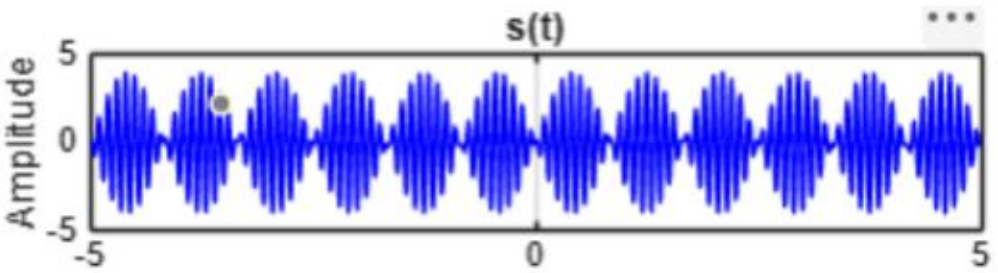


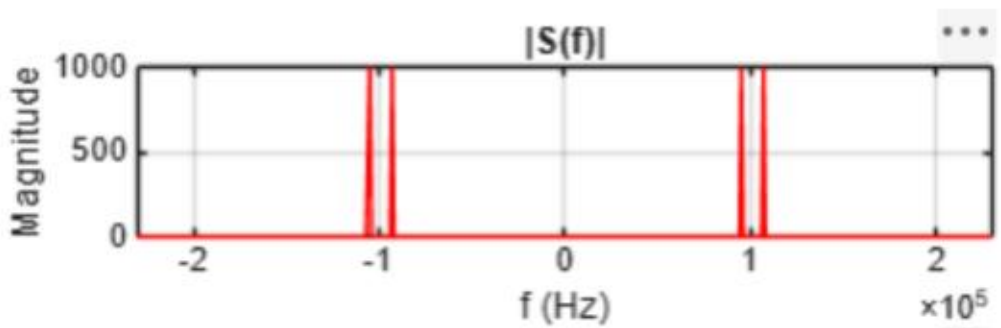
基本公式：s(t) = Ac*(1 + Ka*m(t))*c(t)

- | Ka*m(t) | < 1



重點整理 ▸ DSB-SC (double sideband suppressed carrier)





基本公式：s(t) = m(t)*c(t)

和AM相比

訊號不包含明顯載波分量

較有效率

額外補充

① 變數切記不要重複
(ex : 第一題的座標軸 ~ t1 、第二題座標軸 ~ t2)

② 輸出圖片記得該標的都要標
(ex : 標題名稱、橫/縱軸座標名稱)

③ 善用你的作業和 ctrl+c/v